

(a)

update rule	rotation $\in \text{O}(2)$	stretch $\in \text{Mon}_2$	signed perm. $\in B_2$	generic $\in \text{GL}_2$	gauge group
SGD	9×10^{-17}	0.60	0	0.72	$\text{O}(2)$
Adam, $\varepsilon = 0$	0.41	0.30	0	0.61	B_2
Adam, per-channel η	0.41	1×10^{-16}	0	0.64	Mon_2 (joint)
scaled GD, $\delta = 0.1$	2×10^{-16}	0.015	8×10^{-17}	0.016	$\text{O}(2)$
scaled GD, $\delta = 0$	2×10^{-16}	2×10^{-16}	1×10^{-16}	3×10^{-16}	GL_2
LoRA-Pro	3×10^{-16}	3×10^{-16}	2×10^{-16}	5×10^{-16}	GL_2

(b)

